

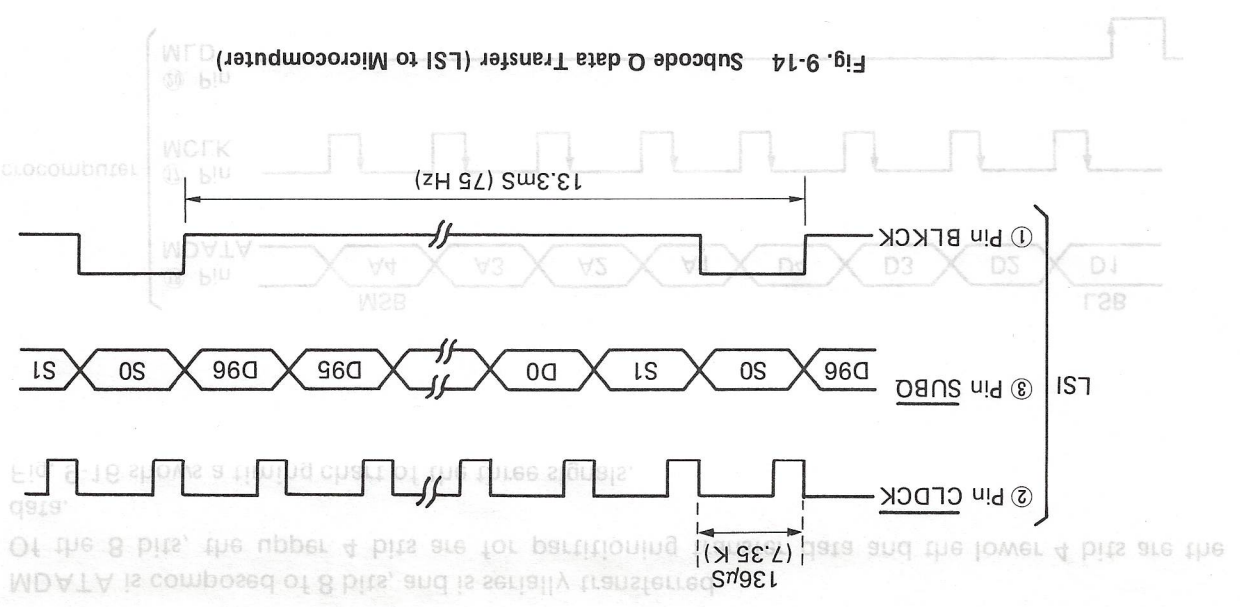
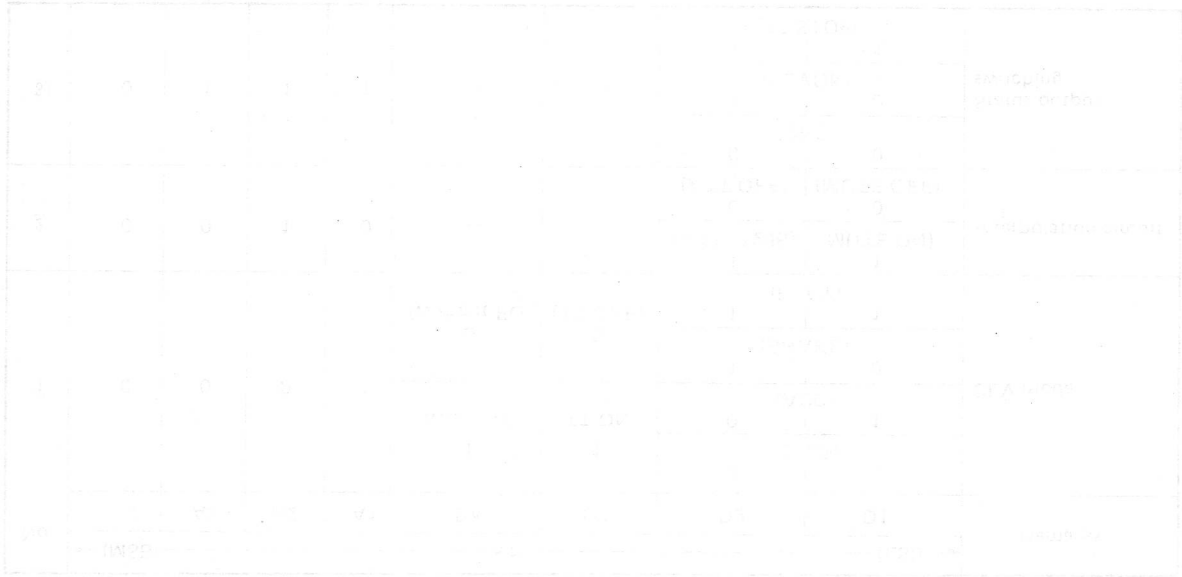


Fig. 9-14 Subcode Q data Transfer (LSI to Microcomputer)

- 2) MLD, MCLK, MDATA signals
- These three signals transfer data from the microcomputer to the LSI.
- (a) MLD : Data loading signal
- (b) MCLK : Data transfer clock signal
- (c) MDATA : Transfer data
- Data is transferred to set the modes of the CLV control and interpolation circuit in the LSI.
- Fig. 9-15 shows MDATA details.

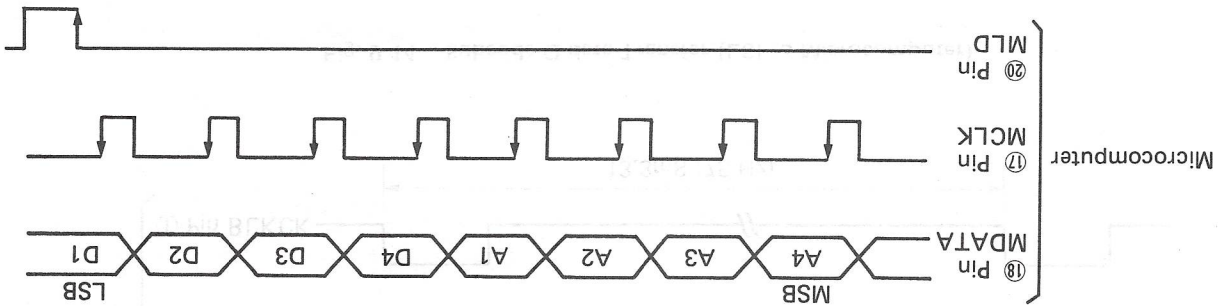
No.	(MSB) → 8 bits → (LSB)							
	A4	A3	A2	A1	D4	D3	D2	D1
①	0	0	0	1	1 (with FG)	1 (TT ON)	0 (STOP)	0 (ACC)
							1 (BRAKE)	1 (PLAY)
							1	1
							0	0
②	0	0	1	0	—	—	1 (ATT-12dB) (MUTE ON)	0 (ATT OFF) (MUTE OFF)
							0	0
③	0	1	1	1	—	—	1 (CLVOK)	1 (TT STOP)
							0	1
Remarks								
	CLV mode							
Interpolation circuit								
	Status output switching							

Fig. 9-15 MDATA Details



STATUS signal is data to be transferred from the LSI to the microcomputer, and its contents are confirmed as necessary by the microcomputer according to the output switching mentioned in Item 3), Fig. 9-15.

Fig. 9-16 MDATA Transfer (Microcomputer to LSI)



MDATA is composed of 8 bits, and is serially transferred. Of the 8 bits, the upper 4 bits are for partitioning transfer data and the lower 4 bits are the data. Fig. 9-16 shows a timing chart of the three signals.

Fig. 9-12 SRF Signal and C0 Signal

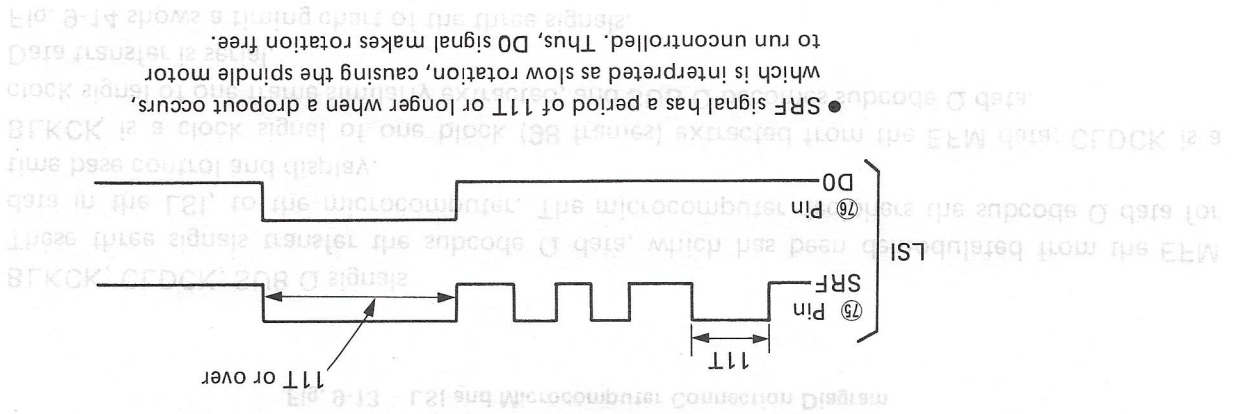
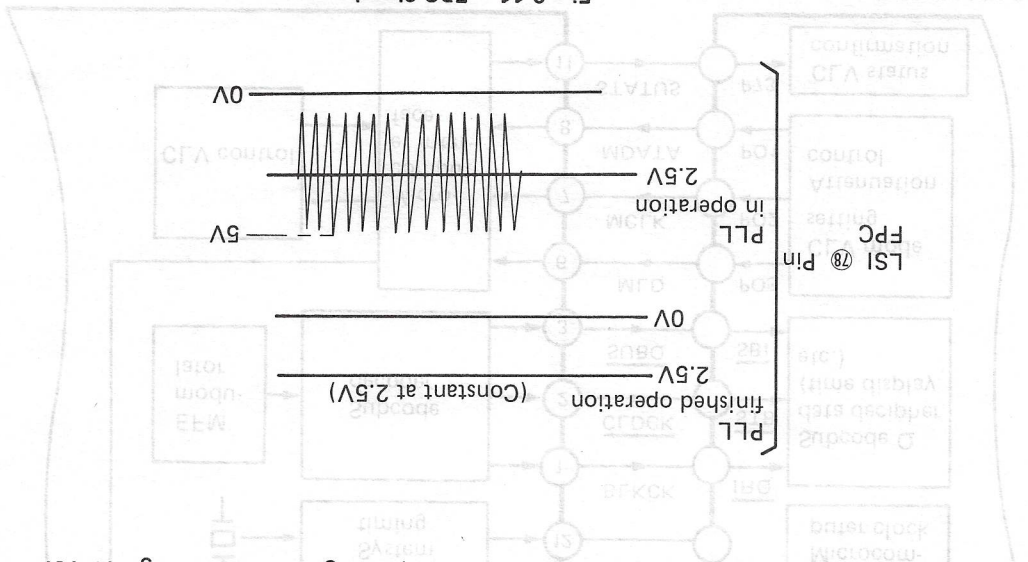


Fig. 9-11 FPC Signal



- D0 signal is a dropout detection signal (dropout detection output of AN8371S) in AN8371S, and is used as a control signal for CLV control in the LSI. The control method is such that, when a dropout is detected, the 11T servo is turned off to allow free rotation. Fig. 9-12 shows SRF signal and D0 signal.
- 3) CNT2 (SLC or TR ON) signal
- CNT2 signal is one of the 4-bit control logic signals of the optical servo IC (AN8370S). This signal remains high until the tracking servo is in the OFF mode (See Table 6-1). This is used to control both ICs as follows.
- (a) The CLV control in the LSI is kept under speed control even during rapid feed by changing the rough servo to the traverse tracking servo mode, and when the tracking servo turns on, the CLV servo is immediately set ready for operation.
- (b) The data slice circuit in AN8371S is so controlled that the slice level is forcibly raised to permit accurate generation of SRF signal as an input signal to the rough servo.

(2) Operations associated with microcomputer
Fig. 9-13 shows an LSI (MN6617) and microcomputer (MN1554CAF) connection diagram. There are four inputs and five outputs from the LSI.

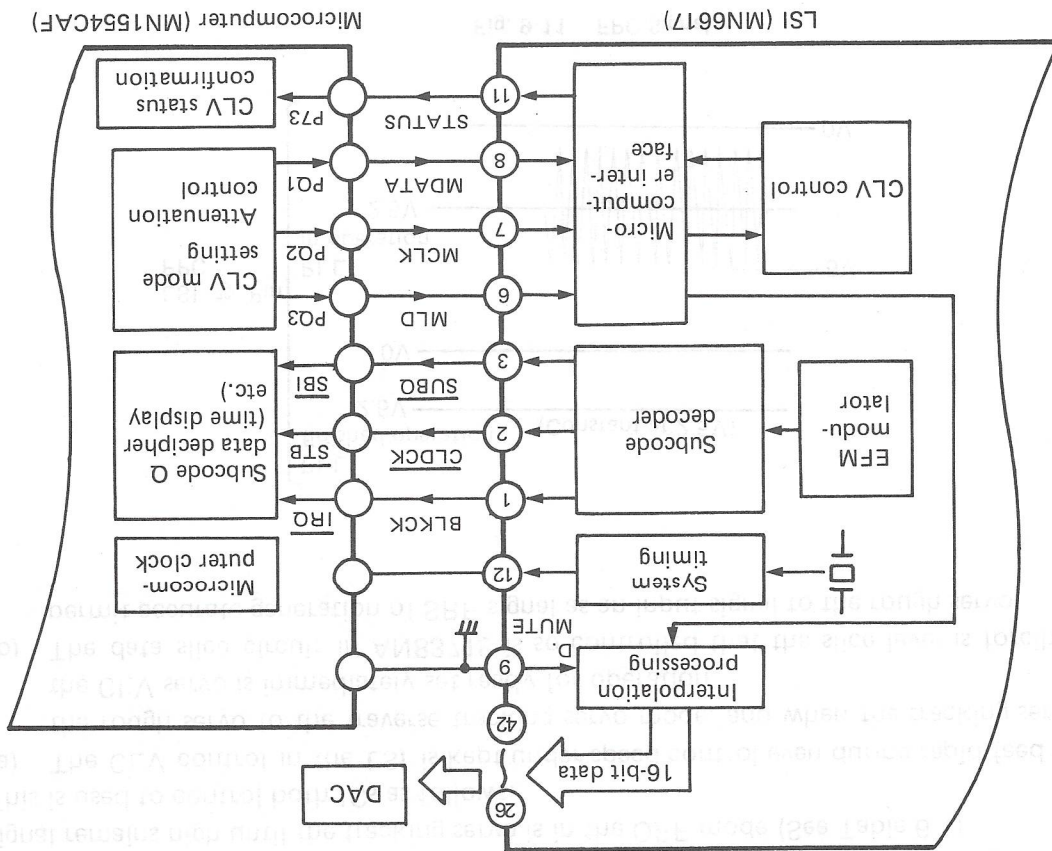


Fig. 9-13 LSI and Microcomputer Connection Diagram

1) BLCK, CLDCK, SUB Q signals
These three signals transfer the subcode Q data, which has been demodulated from the EFM data in the LSI, to the microcomputer. The microcomputer deciphers the subcode Q data for time base control and display.
BLCK is a clock signal of one block (98 frames) extracted from the EFM data; CLDCK is a clock signal of one frame similarly extracted, and SUB Q becomes subcode Q data.
Data transfer is serial.
Fig. 9-14 shows a timing chart of the three signals.

Fig. 9-9 shows an LSI block diagram.

- 1) EFM demodulation, subcode demodulation
In EFM demodulation, an extraction clock PCK is used to latch EFM signal every 14 bits to return it to the original 8-bit data.
The frame clock of 7.35 kHz in EFM signal is output from pin ② (CLDCK) by synchronized detection.
In subcode demodulation, the synchronized signal in the 14 bits preceding EFM demodulation is detected, and subcode Q data is output to pin ③ (SUBQ) and emphasis/de-emphasis signal to pin ④ (DEMPHA) (high in case of emphasis signal).
Also, a subcode Q block clock of 75 Hz is output from pin 1 (BLKCK).

- 2) C1, C2 unscramble, deinterleave
Time base processing (data rearrangement) is used during recording to deal with various burst errors and random errors that occur through the recording and reproducing systems.
The data thus rearranged is returned to the original state by an operation opposite to that of recording, using an external RAM (16K).
(a) Pins 43 to 50 (D0 to D7)
(b) Pin ⑤ (RAM WE)
(c) Pin ⑥ (RAM OE)
(d) Pins ⑤ to ⑥ (RAMA0 to RAMA10) : RAM address signals
3) Error detection, correction (C1, C2 decoders)
The C1 decoder detects errors in data subsequent to C1 unscramble and corrects them, using parity P signal. The C2 decoder detects errors in data subsequent to deinterleave and corrects them, using parity Q signal.

- 4) System timing
Xtal signal (16.9344 MHz) that is connected to pin ⑩ (X2) and pin ⑪ (X1) is divided to generate a clock for signal processing in the LSI and a timing clock signal for external circuits.
(a) Pin ⑫ (SMCK)
Signal output of 4.23 MHz
(b) Pin 19 (LDG), pin 20 (RDG)
Signal output of 44.1 kHz used as a deglitch signal for audio circuit

5) CLV servo
The CLV servo circuit is described in 9-3 (3).

9-3 LSI and Associated Operations of Peripheral Circuits

(1) Operations associated with AN8371S

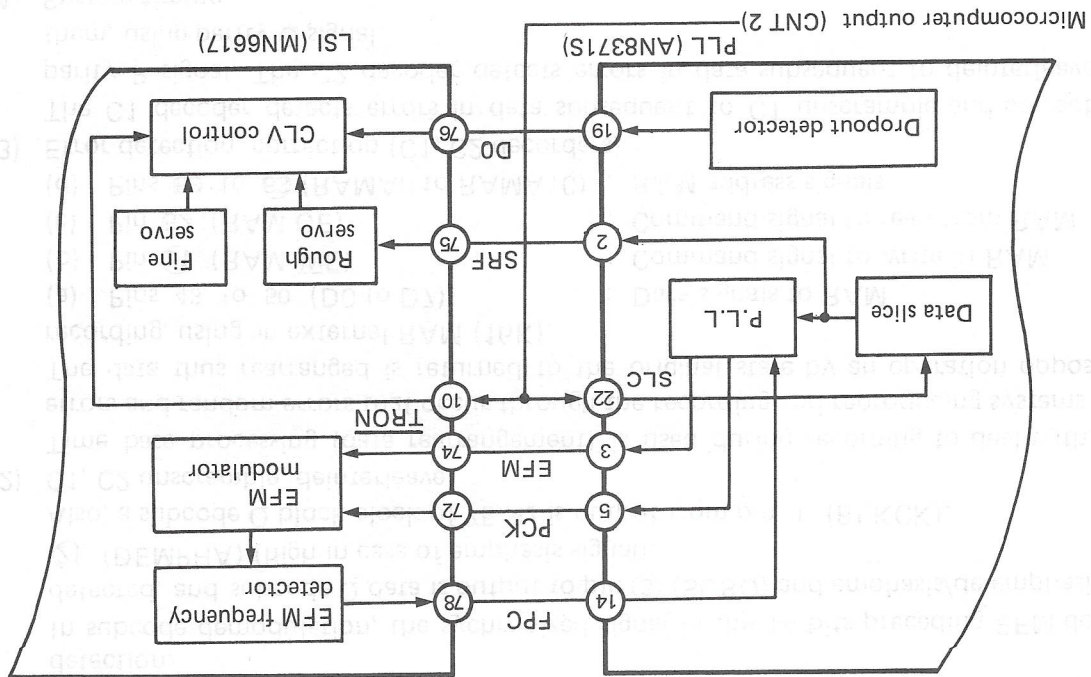


Fig. 9-10 AN8371S and LSI Connection Diagram

Fig. 9-10 shows an LSI (MN6617) and PLL IC (AN8371S) connection diagram. There are four inputs and one output as viewed from the LSI, and the output (CNT2) from the microcomputer is input to both ICs.

1) PCK, EFM, FPC signals

PCK and EFM signals are as described in Section 6.

FPC signal results from measuring the pulse width of EFM signal (3T to 11T) with PCK signal (1T).

If EFM signal is a signal of 2T or less, a feedback operation takes place to raise the VCO frequency of the PLL. If EFM signal is 12T or over, the VCO frequency is lowered.

During play (PLL has completed operating), pin 78 of the LSI is constant at 2.5V. If the PLL is still operating, a control pulse of $2.5V \pm 2.5V$ is output.

2) SRF, D0 signals

SRF signal is generated by data-slicing RF signal (data slice output of AN8371S), and is used as an input signal for the CLV rough servo (11T servo) in the LSI. The CLV servo is described later.

9-2 Reproducing System Signal Processing

(1) Decoder

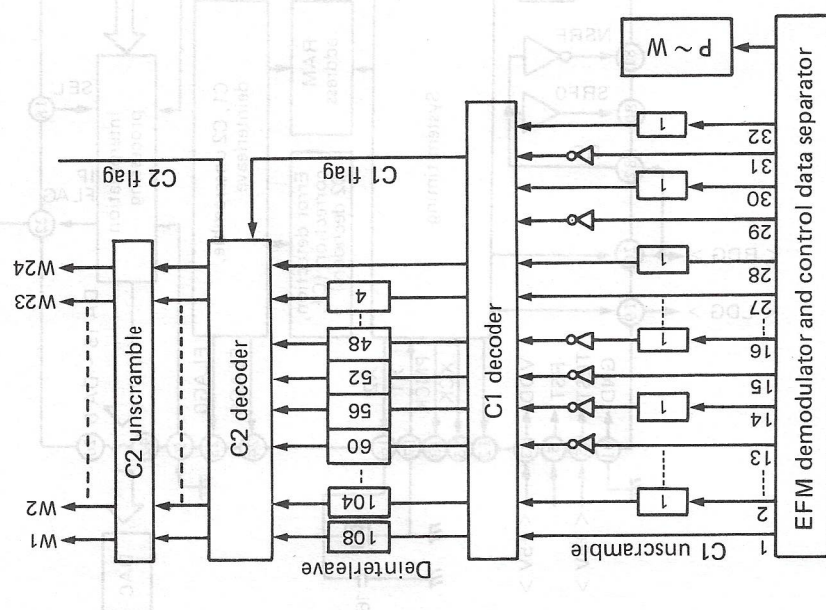


Fig. 9-8 Reproducing System Signal Processing

A reproducing system signal processing block diagram is shown in Fig. 9-8. The flow of reproducing system signal processing is just opposite to the flow of recording system signal processing. Here, decoders are error detection and correction are used in place of the C1 and C2 encoders that are used for parity generation. Both the C1 and C2 decoders are capable of error detection and correction. If data is incorrectable in the C1 decoder, C1 flag is attached to that data, and after deinterleaving, the data is sent to the C2 decoder for accurate correction. If data is incorrectable in the C2 decoder, an average value is obtained from the preceding and next data after C2 unscramble, followed by interpolation processing.

(2) LSI signal processing

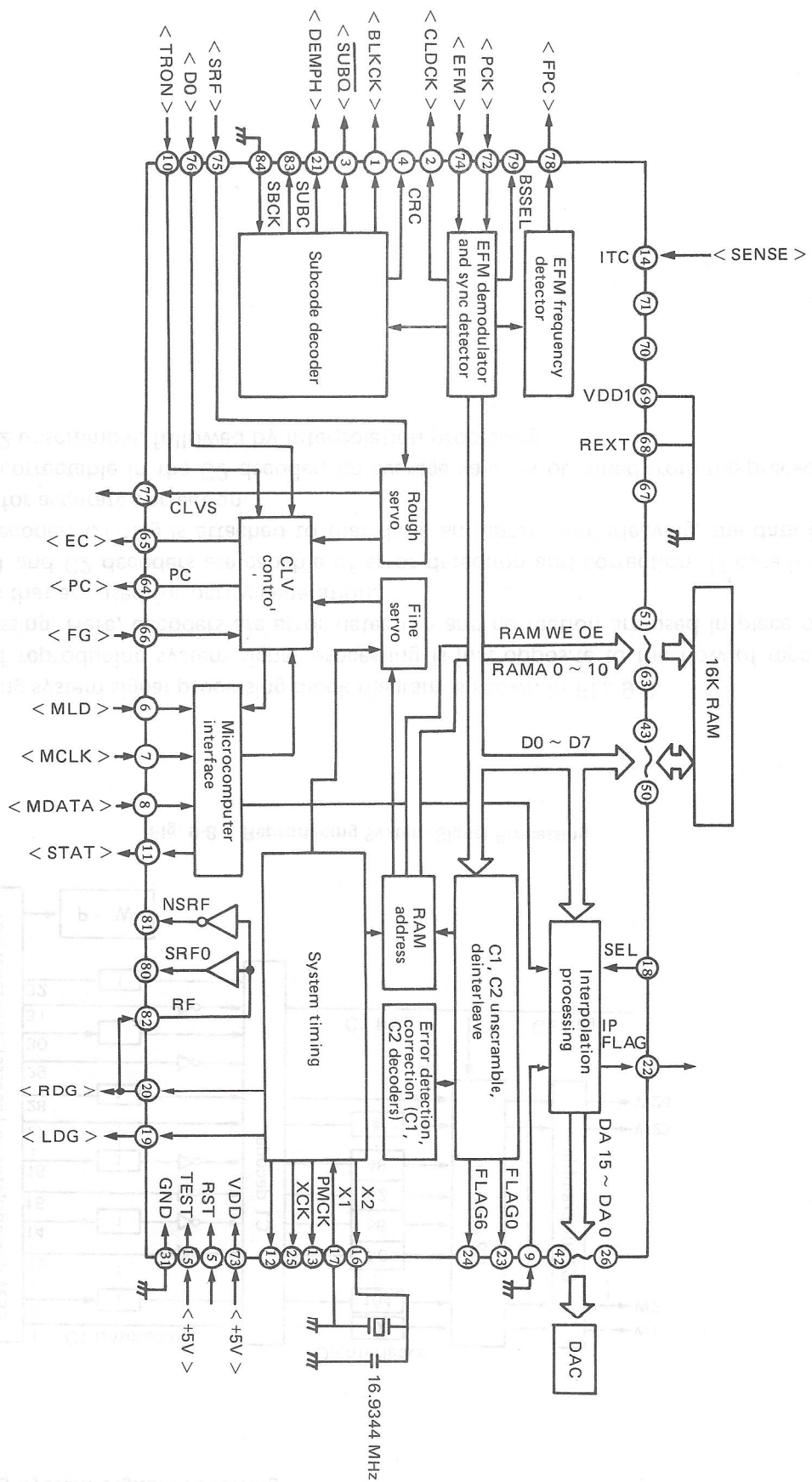


Fig. 9-9 LSI Block Diagram

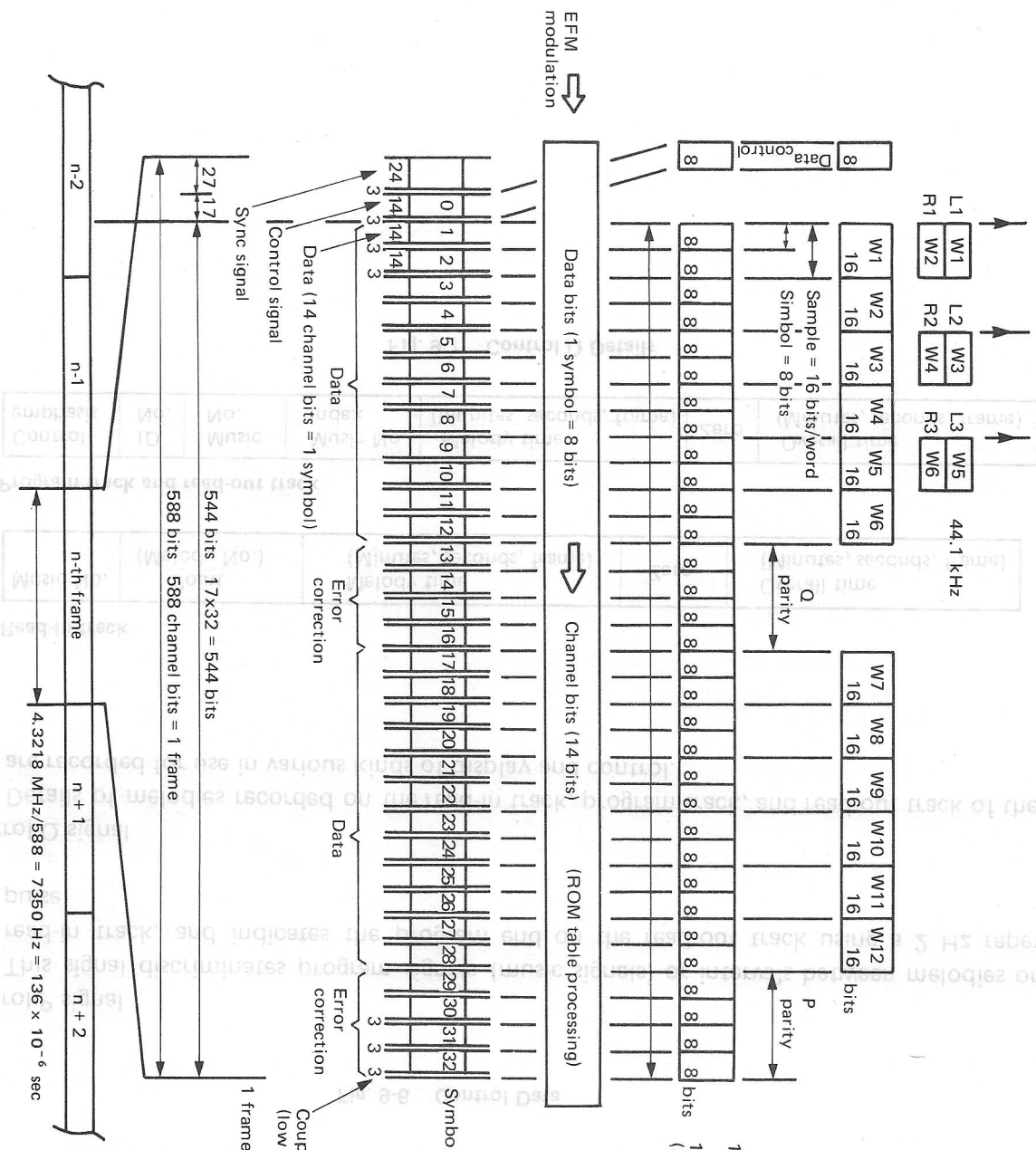


Fig. 9-5 CD Signal Format

* 3 bits between 14 bits call the coupled bits and when they are observed through the data, they serve as connectors to maintain the EFM standards.

12 samples, 16 data
4 parties
16x16 = 256 bits
(8 x 32)

Coupled bits
(low range suppression)
1 frame = 12 samples + 4 parties
+ control signal + Sync signal

Control data is recorded in one symbol (8 bits) in one frame. The 8 bits are named P, Q, R, S, T, U, V, and W in this order. Actually, however, only bits P and Q of the 8 bits are used.

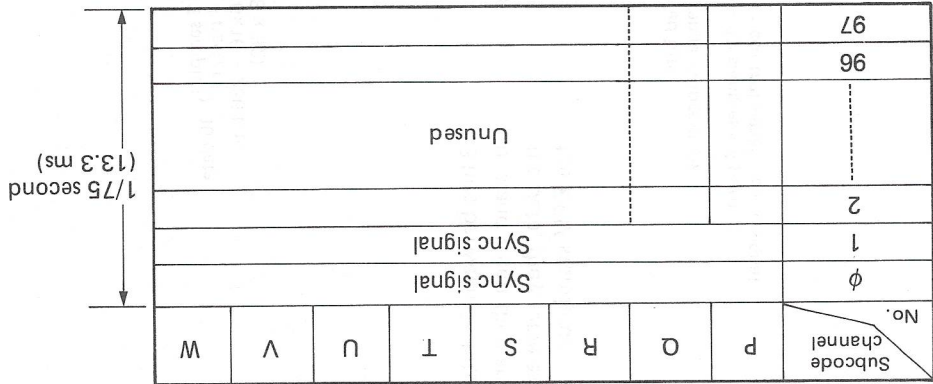


Fig. 9-6 Control Data

Control P signal
This signal discriminates program signals (music signals) or intervals between melodies on the read-in track, and indicates the program end on the read-out track using a 2 Hz repetitive pulse.

Control Q signal
Details of melodies recorded on the read-in track, program track, and read-out track of the disk are recorded for use in various kinds of display and control.

Music No.	Point (Melody No.)	Melody time (Minutes, seconds, frame)	Zero	Overall time (Minutes, seconds, frame)
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Read-in track

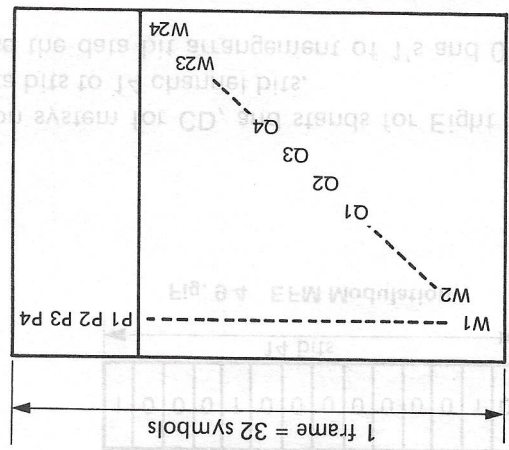
Control emphasis	ID No.	Music No.	Music No. index	Melody time (Minutes, seconds, frame)	Zero	Overall time (Minutes, seconds, frame)
------------------	--------	-----------	-----------------	---------------------------------------	------	--

Program track and read-out track

Fig. 9-7 Control Q Details

Fig. 9-3 shows the process of up to the recording signal format. Therefore, signals to be recorded have definitions with 3T to 11T (T meaning 1 clock after EFM) for correspondence to 8 data patterns. EFM selects bit patterns in which 5 to 10 0's are present between two 1's from among 2¹⁴ combinations. 8 bits has 256 combinations, while 14 bits has 16,384 combinations (2¹⁴). Therefore, the DC component is as small as possible. Therefore, signals to be recorded are reversed at appropriate intervals very small diameter. Conversely, it is necessary to reduce the laser beam to a diameter of 1.5 μm and 0.5 μm and 0.5 μm and 0.5 μm are added to the 24 symbols arranged diagonally.

Fig. 9-3 CIRC Code Composition



Here, the symbol data count in a frame is 32 in total. (See Fig. 9-3.)

- end.
- 6) The C1 encoder generates 4-symbol parity signals P1 to P4 from the 28-symbol data, which have been rearranged by interleaving, using reed solomon code, and arranges them at the frame over the range of 4 frames minimum to 108 frames maximum.
- As shown in the figure, the interleave delays the 28-symbol data, including the Q parity signals, Next is the interleave.
- 5) The symbol data count in one frame is 28 in total.
- center.
- Q1 to Q4 from the 24-symbol data using reed solomon code, and arranges them in the frame
- 4) After the C2 scramble, the data enter the C2 encoder, which generates 4-symbol parity signals in C2 scramble, the data are delayed over two frames alternately in units of sample data.
- 3) Data W1 to W24 are first scrambled by C2. Scramble may be considered to be a small interleave. other words, 24-symbol data constitutes a frame. W1 to W24 shown in Fig. 9-1 correspond to 12 samples of data (six each for the right and left) are taken as a frame for signal processing. In

- (2) EFM modulation
- 7) The C1 scramble delays only the even-numbered symbols of the 32-symbol data in a frame to the next frame.
 - 8) Control data P to W are otherwise called subcodes or users bits, containing time information for random access or display and consisting of a single symbol (8 bits).
 - These signals are arranged at the head of a frame.
 - Entry of these control signals increases the symbol data count in a frame to 33 in total.
 - 9) Finally, the above 1-frame data are EFM-modulated, a frame synchronizing signal is added, and the data are recorded on the disk.
 - EFM modulation is described in detail in the next paragraph.
 - All these are intended for improving the correction capacity.
 - This system is called CIRC (Cross Interleave Reed Solomon Code).

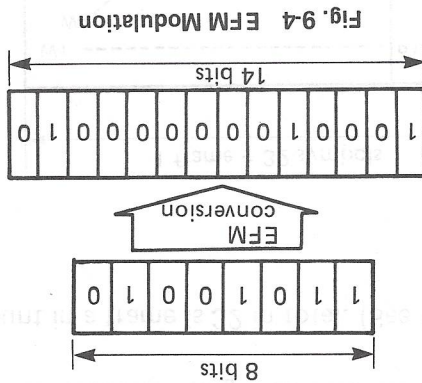


Fig. 9-4 EFM Modulation

EFM is a new modulation system for CD, and stands for Eight to Fourteenth Modulation meaning modulating 8 symbol data bits to 14 channel bits.

EFM is necessary because the data bit arrangement of 1's and 0's cannot be defined if symbol data remains as such.

If 1's or 0's continue so long, data will be a direct current so that self-clocks cannot be generated by the tracking servo (using pits) and reproducing system.

Conversely, if 1's and 0's are frequently reversed, it will be necessary to reduce the laser beam to a very small diameter.

Therefore, signals to be recorded must be such that 1's and 0's are reversed at appropriate intervals and that the DC component is as small as possible.

8 bits has 256 combinations (2^8), while 14 bits has 16,384 combinations (2^{14}).

EFM selects bit patterns, in which 2 to 10 0's are present between two 1's, from among 2^{14} combinations for correspondence to 2^8 data patterns.

Therefore, signals to be recorded have definitions with 3T to 11T (T meaning 1, 1 clock after EFM = 4.32 MHz).

Fig. 9-5 shows the process of up to the recording signal format.

8-2 PCK Signal

PCK signal is a clock signal synchronized with EFM signal. EFM signal has some jitter components until it is written into the memory. Therefore, a clock signal synchronized with EFM signal is necessary in order for the digital signal processing circuit to accurately receive data. PCK signal is generated from an EFM signal input by the PLL circuit. Fig. 8-3 shows a timing chart of EFM signal and PCK signal.

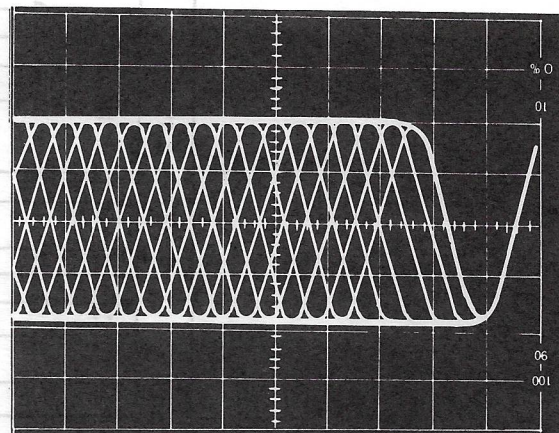


Fig. 8-2 RF Signal

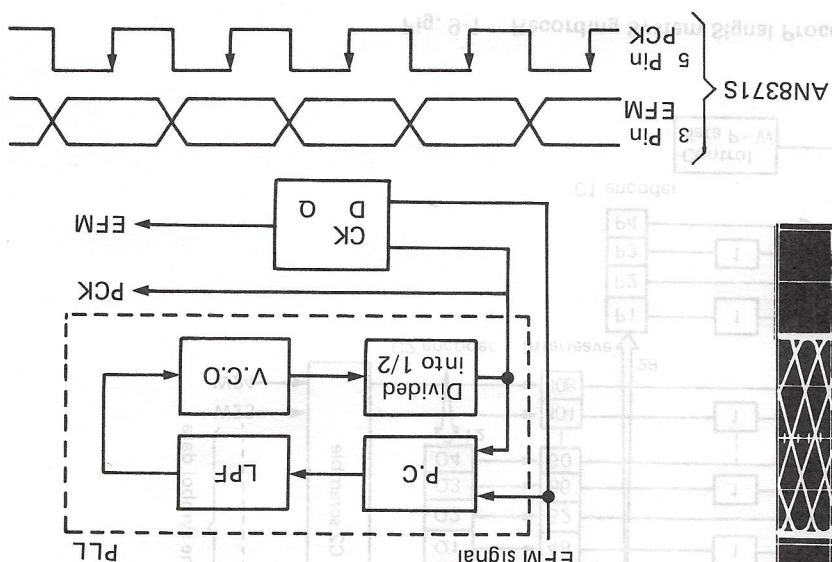


Fig. 8-3 EFM Signal and PCK Signal

8-3 Other Signals

A part from EFM signal and PCK signal, AN8371S outputs SRF signal and D0 signal, and receives input signals of FPC and SLC. SRF signal is generated by data-slicing RF signal. D0 signal is a dropout detecting signal for RF signal (high at dropout). Both are input to the CLV control in LSI (MN6617). FPC signal is a VCO control signal for the PLL circuit that is fed back from the LSI, and SLC signal is a slice level control signal from the microcomputer (MN1554CAC). Their functions are described in 9-3.

9. MN6617 CIRCUIT OPERATION

Before beginning a circuit operation description, the signal processing of the recording system is explained.

9-1 Recording System Signal Processing (1) Encoder

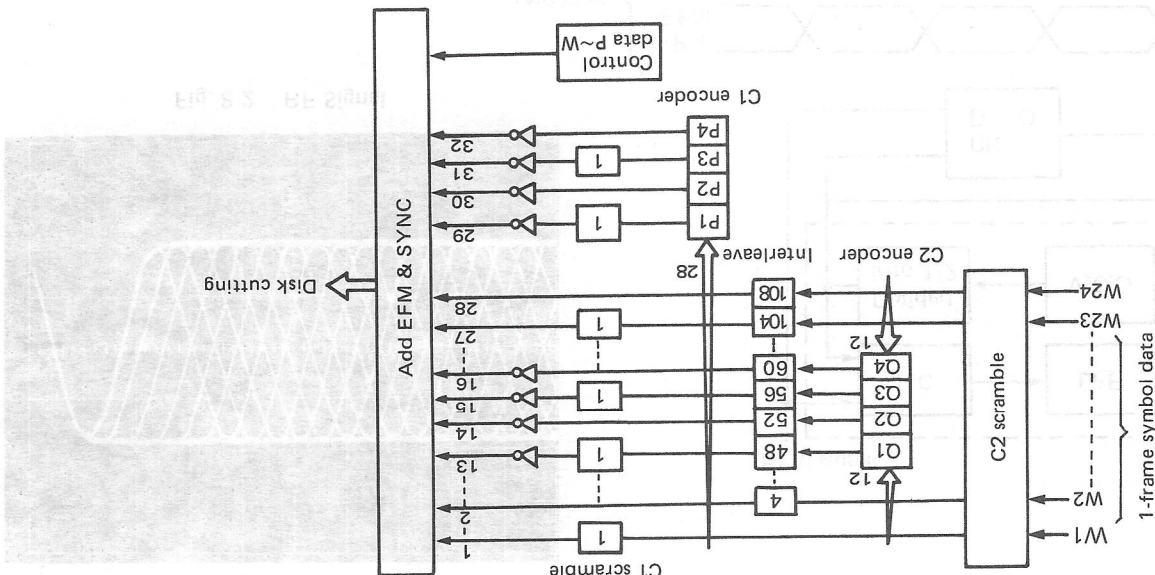


Fig. 9-1 Recording System Signal Processing

Fig. 9-1 shows a block diagram of CD recording system signal processing. A brief description of CD recording system signal processing is given in the following.

- 1) The CD sampling frequency is 44.1 kHz, and quantizing involves A-D conversion to 16 bits per sample.

The 16 bits are divided into two parts, upper 8 bits and lower 8 bits.

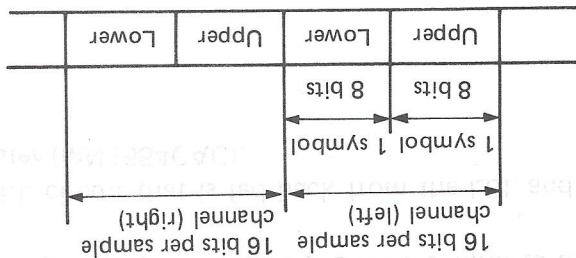


Fig. 9-2 Symbol Data and Other Than Data

7-1 Traverse Detector Circuit

The tracking error signal corresponds to the eccentric component of the disk as viewed within the range of a single disk revolution. In a little wider range, however, a DC voltage (servo error voltage) is generated as shown in Fig. 7-2. This DC voltage drives the traverse mechanism.

The traverse detector circuit referred to here covers from the tracking coil output to the lowpass filter (pin 1 of IC601) output. Pin 8 of IC601 outputs forced drive signals, that is, forced forward H and reverse L, by micro-computer commands.

Pin 7 of IC601 is set to a reference voltage of +5V.

7-2 Mode Select Circuit

The multiplexer (IC603) is used for selecting three inputs in mode selection.

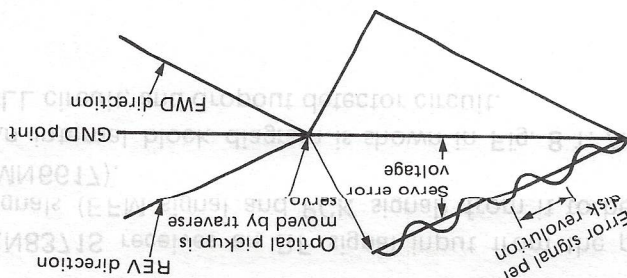
- 1) Servo error voltage selection
The output of pin 1, IC601 is applied to the drive circuit when pin 10 and pin 9 of IC603 are high.
- 2) Brake selection by reference voltage
The output of pin 7, IC601 is applied to the drive circuit when pin 10 of IC603 is low and pin 9 of IC603 is high.
- 3) Forced forward/reverse selection
The output of pin 8, IC601 is applied to the drive circuit when pin 9 of IC603 is high.

The input to pin 9 of IC603 is controlled by CNT1 to CNT3 of the microcomputer via the NOR circuit. Thus, pin 9 goes high when all of CNT1 to CNT3 are low (during disk play), or goes low when one of CNT1 to CNT3 is high (other than during disk play).
The input to pin 10 of IC603 is controlled by CLVS signal 77 of LSI (IC501). Thus, pin 10 goes high when T11 is OK.

7-3 Traverse Drive Circuit

The traverse drive circuit consisting of IC602, Q606 to Q609 drives the traverse motor using the reference voltage applied to pin 3 of IC602 according to the polarity corresponding to the voltage difference applied to pin 2 and the voltage difference.

Fig. 7-2 Traverse Servo Voltage



8. AN8371S CIRCUIT OPERATION

AN8371S receives an RF signal input from the preceding optical servo IC (AN8370S), and generates signals (EFM signal and PCK signal) from it to be input to the subsequent digital signal processing LSI (MN6617). An internal block diagram is shown in Fig. 8-1. AN8371S consists of an AGC circuit, data slice circuit, PLL circuit, and dropout detector circuit.

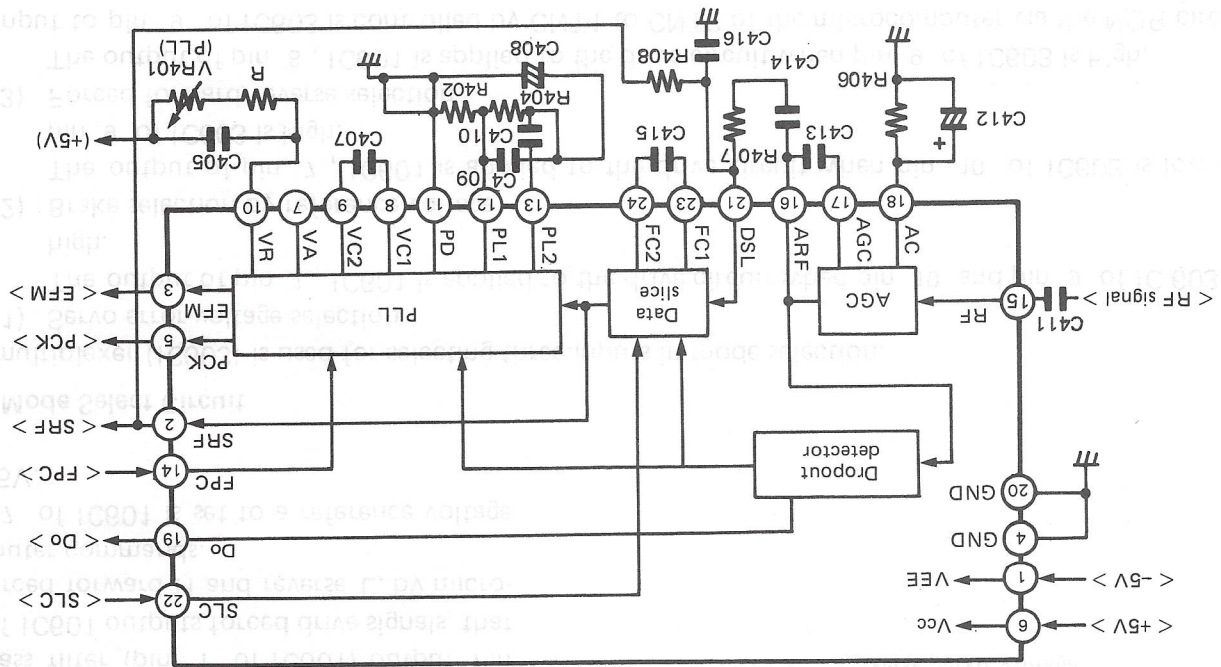


Fig. 8-1 AN8371S Block Diagram

8-1 EFM Signal

RF signal is a continuous signal as shown in Fig. 8-2. Therefore, it cannot be processed as a digital signal unless it is shaped into a digital signal. This waveform shaping is performed by the data slice circuit, whose output signal is EFM signal.

6-6 Focus Tracking Drive Circuit

Fig. 6-14 shows the focus tracking servo drive circuit. Both servos have the same drive circuit except that the focus servo has a search circuit it and that the tracking servo has a kick hold circuit.

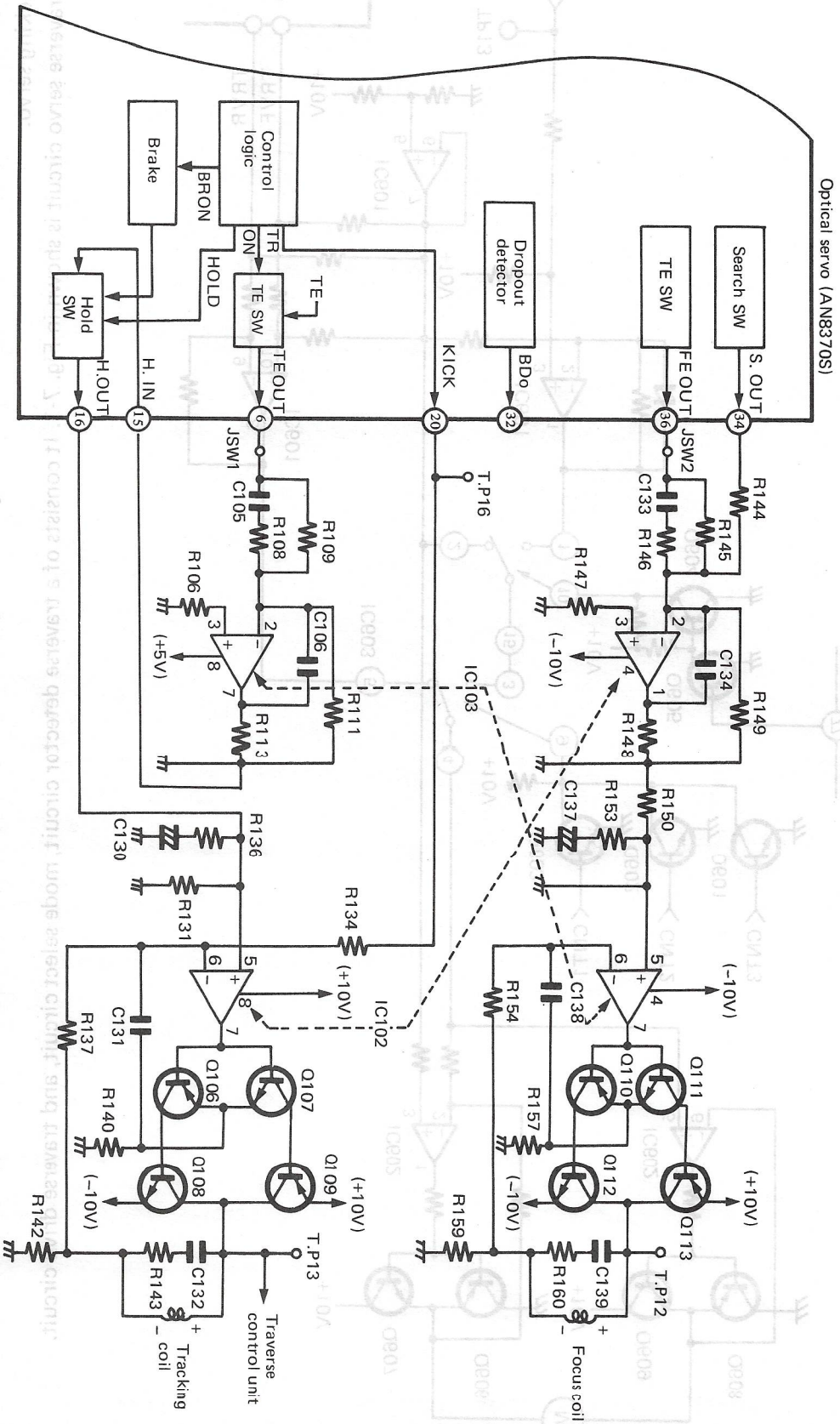


Fig. 6-14 Focus Tracking Drive Circuit

7. TRAVERSE SERVO CIRCUIT OPERATION

The traverse servo detects the inclination of the objective lens, moves the optical pickup, and thus keeps the objective lens within the range of tracking servo.

The traverse servo circuit is shown in Fig. 7-1. It consists of a traverse detector circuit, mode select circuit, and traverse drive circuit.

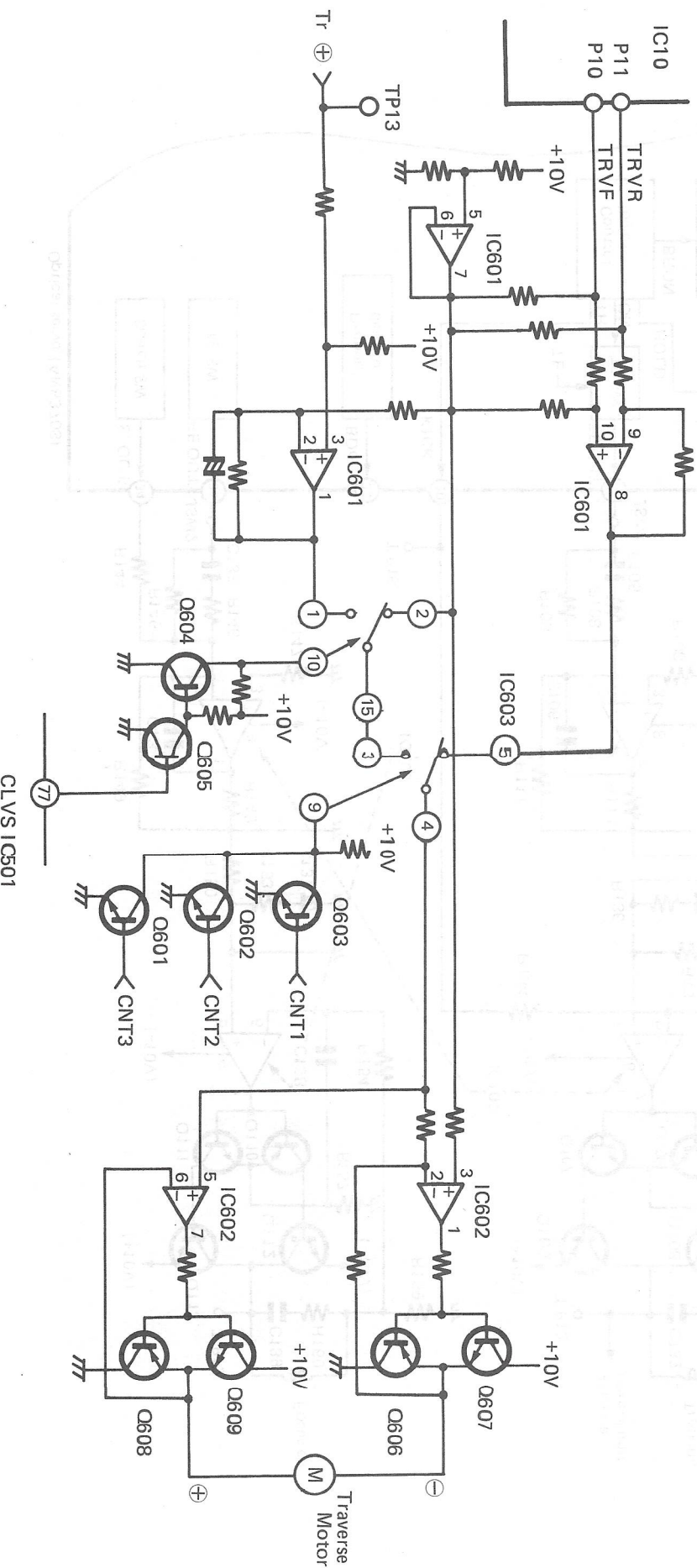


Fig. 7-1 Traverse Servo Circuit

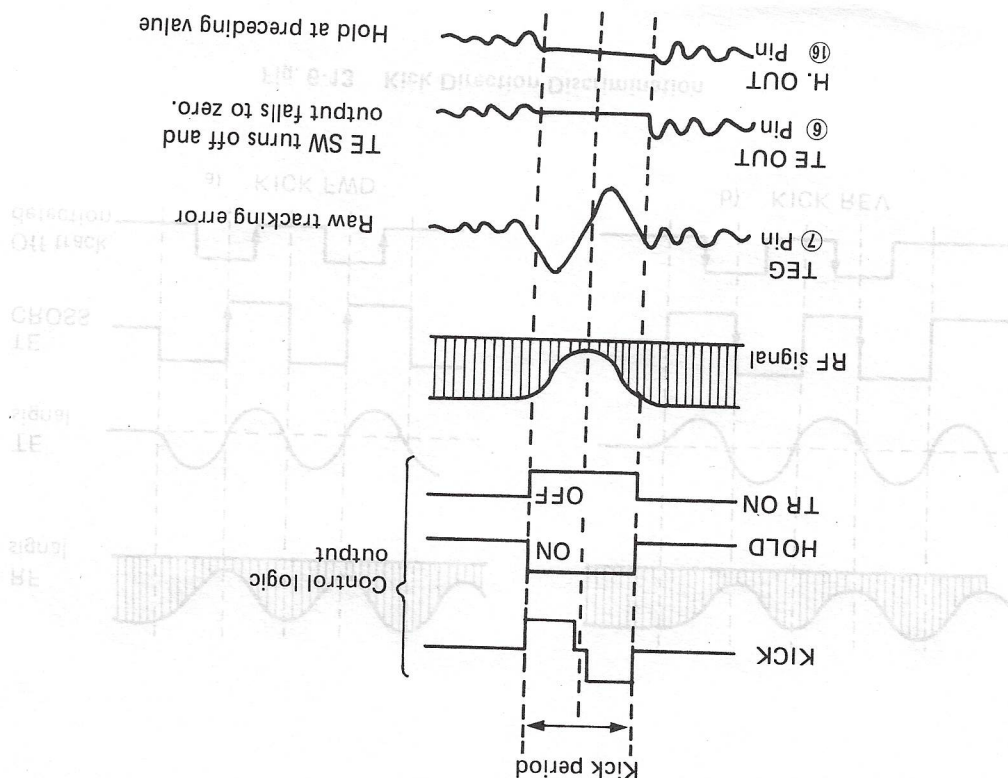
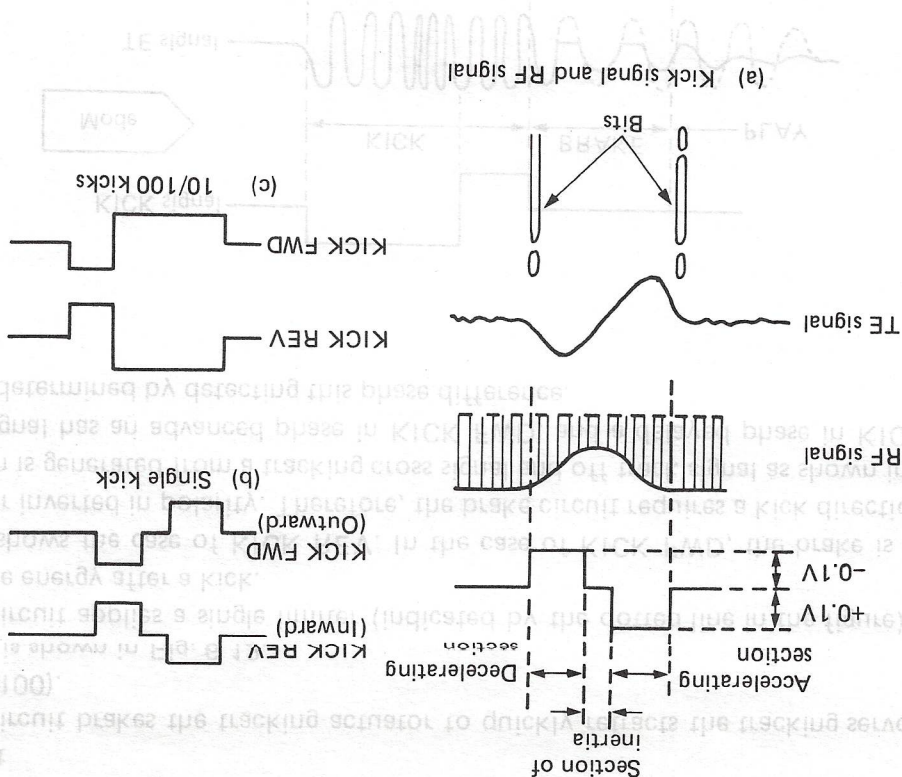


Fig. 6-10 Kick Signal Waveforms



The brake circuit brakes the tracking servo to quickly retract the tracking servo after rapid feed or kick (10/100). The process is shown in Fig. 6-12. The brake circuit applies a single limiter (indicated by the dotted line in the figure) to the TE signal to absorb the energy after a kick. The figure shows the case of KICK FWD, the brake is applied with the single limiter inverted in polarity. Therefore, the brake circuit requires a kick direction discriminating signal, which is generated from a tracking cross signal and off track signal as shown in Fig. 6-13. The cross signal has an advanced phase in KICK FWD, and a delayed phase in KICK REV. Braking direction is determined by detecting this phase difference.

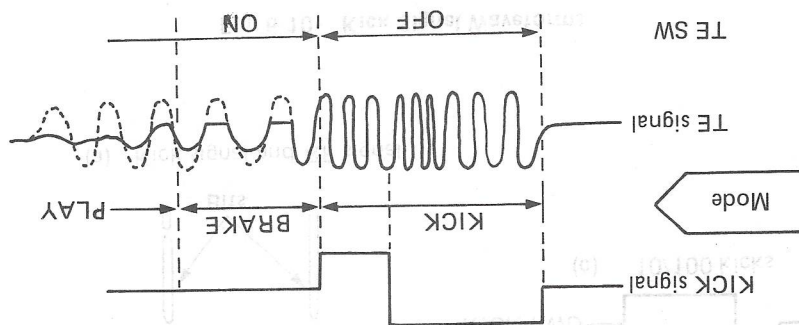


Fig. 6-12 Brake Timing

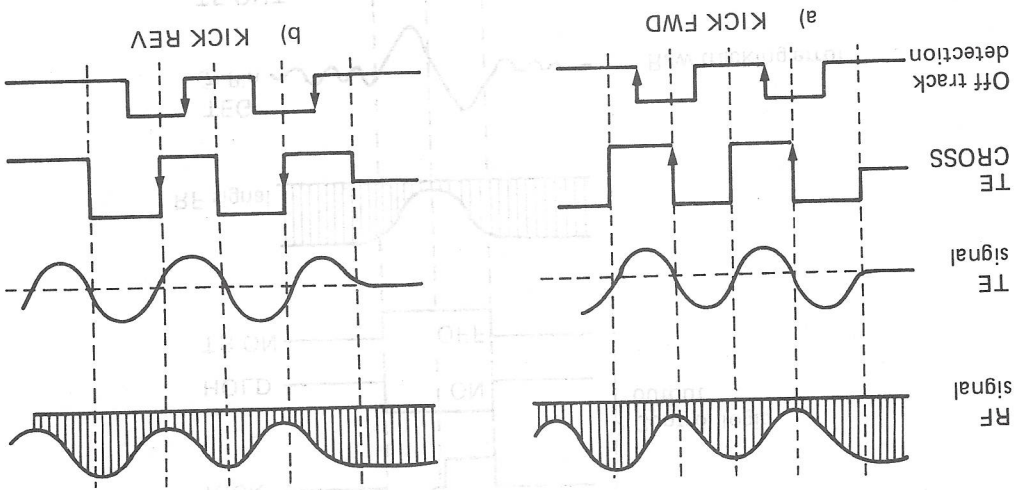


Fig. 6-13 Kick Direction Discrimination